

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-7: Image Enhancement in the Spatial Domain

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-3)

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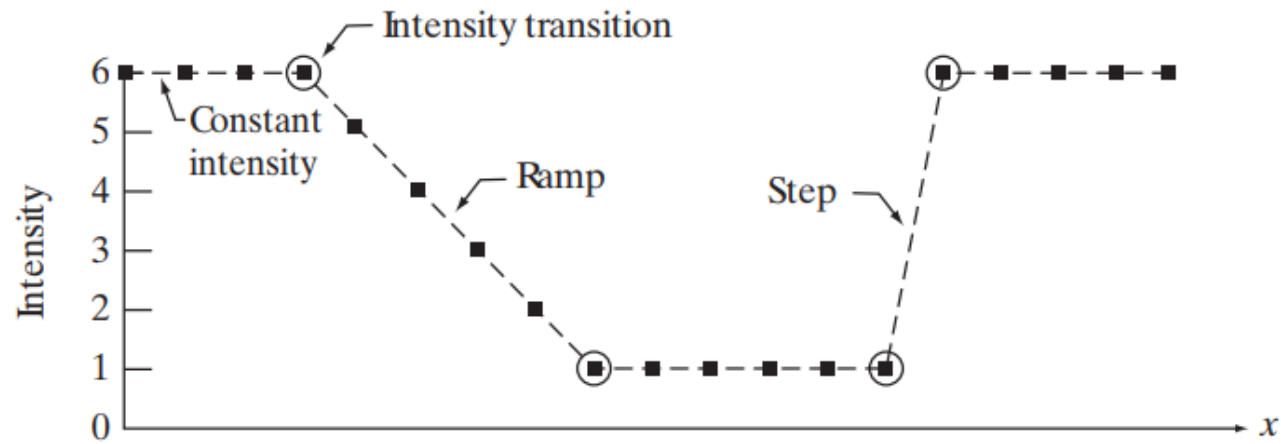
7.6 Use of Second Derivatives for Enhancement–The Laplacian

The approach basically consists of defining a discrete formulation of the second-order derivative and then constructing a filter mask based on that formulation.

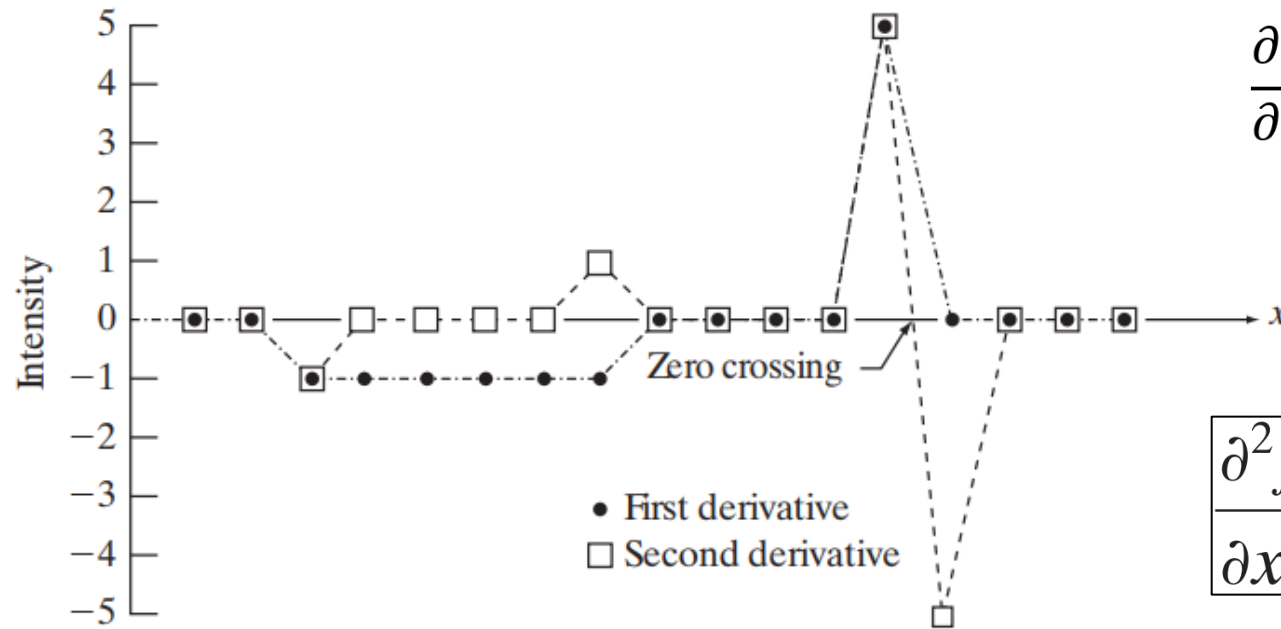
We are interested in isotropic filters, whose response is independent of the direction of the discontinuities in the image to which the filter is applied.

Isotropic means?

Isotropic filters are rotation invariant, in the sense that rotating the image and then applying the filter gives the same result as applying the filter to the image first and then rotating the result.



Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	$\rightarrow x$
1st derivative	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0	5	0	0	0	0	0	
2nd derivative	0	0	-1	0	0	0	0	0	1	0	0	0	0	5	-5	0	0	0	0	



$$\frac{\partial f}{\partial x} = f(x + 1) - f(x)$$

$$\frac{\partial^2 f}{\partial x^2} = f[f(x + 1) - f(x)] - [f(x) - f(x - 1)]$$

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

Development of the method

The simplest isotropic derivative operator is the Laplacian, which, for a function (image) $f(x, y)$ of two variables, is defined as

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.$$

Because derivatives of any order are linear operations, the Laplacian is a linear operator.

$$\frac{\partial^2 f}{\partial x^2} = f(x + 1) + f(x - 1) - 2f(x)$$

The partial second-order derivative in the x-direction

$$\frac{\partial^2 f}{\partial^2 x^2} = f(x + 1, y) + f(x - 1, y) - 2f(x, y)$$

and, similarly in the y-direction, as

$$\frac{\partial^2 f}{\partial^2 y^2} = f(x, y + 1) + f(x, y - 1) - 2f(x, y)$$

The digital implementation of the two-dimensional Laplacian is obtained by summing these two components:

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

$$\nabla^2 f = [f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1)] - 4f(x, y).$$

This equation can be implemented using the mask

0	1	0
1	-4	1
0	1	0

The diagonal directions can be incorporated in the definition of the digital Laplacian by adding two more terms

1	1	1
1	-8	1
1	1	1

Two other implementations of the Laplacian

0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

This highlights edges and high-frequency components.

Use the Laplacian for image enhancement is as follows:

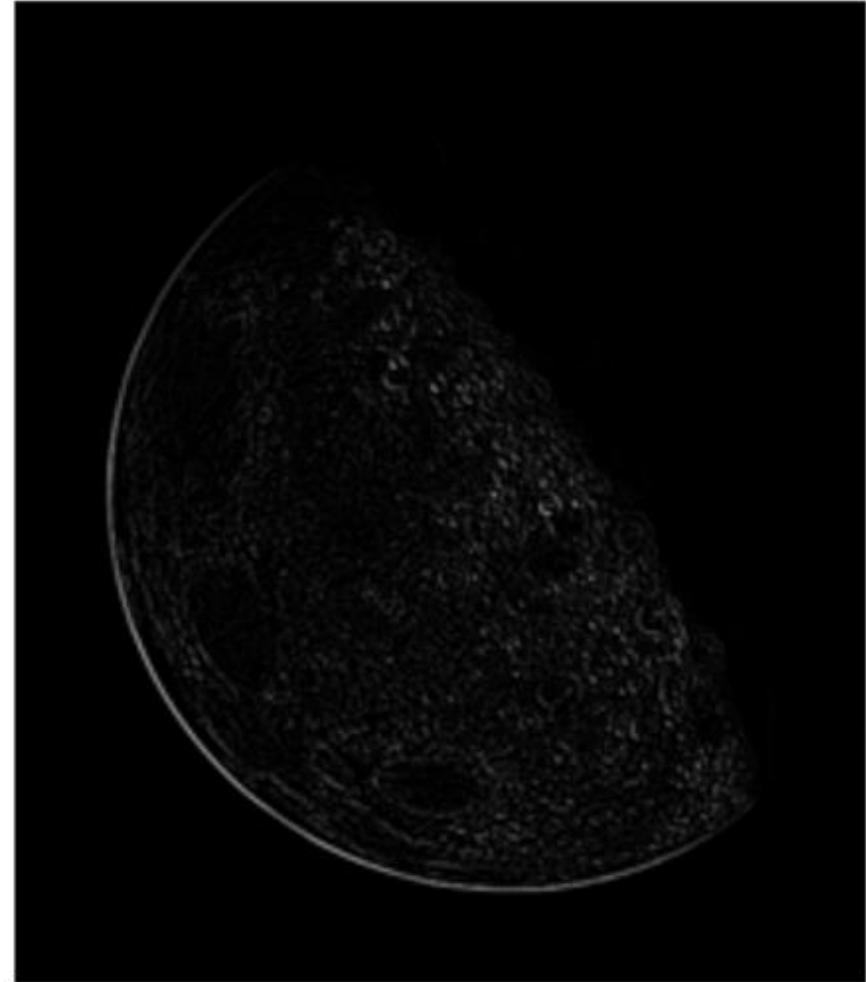
$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is positive.} \end{cases}$$

0	1	0
1	-4	1
0	1	0

0	-1	0
-1	4	-1
0	-1	0

$$\nabla^2 f = [f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1)] - 4f(x, y).$$

Example: Image of the North Pole of the moon

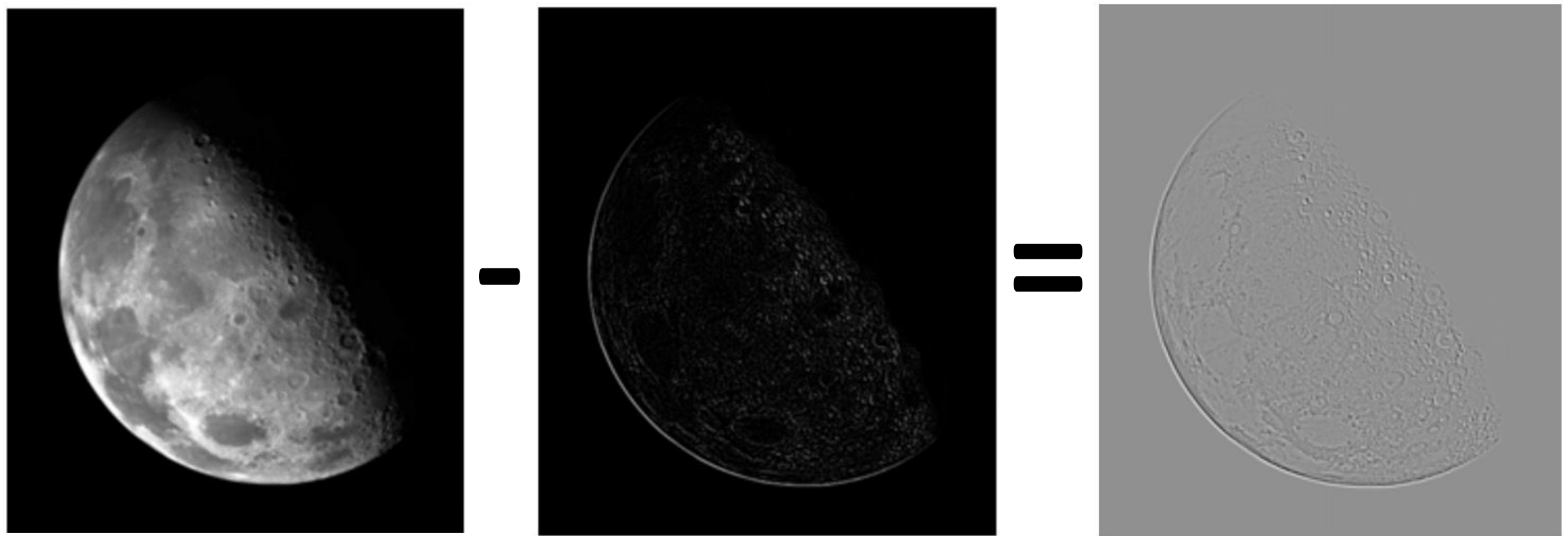


First computing the Laplacian-filtered image

then subtracting it from the original image.

$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) \\ f(x, y) + \nabla^2 f(x, y) \end{cases}$$

Laplacian
filtered image



$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) \\ f(x, y) + \nabla^2 f(x, y) \end{cases}$$

Note that the dominant features of the image are edges and sharp gray-level discontinuities of various gray-level values.

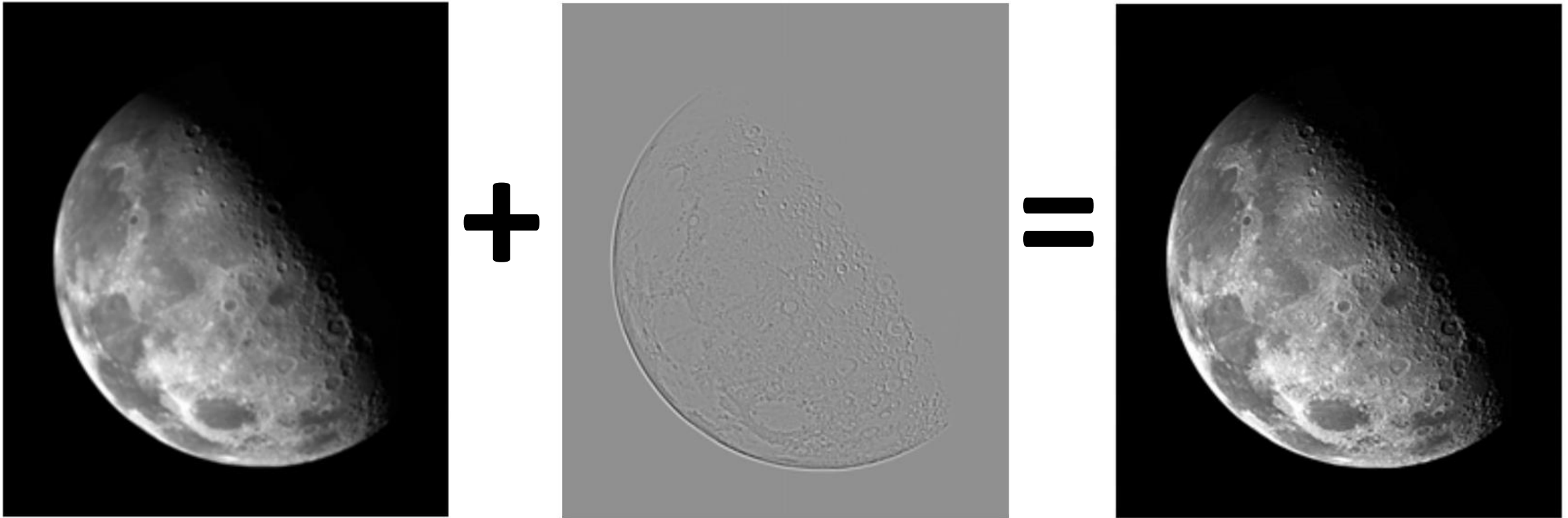


$g(x, y)$

Add the Laplacian-filtered image to the original image to enhance the details:

$$I_{\text{enhanced}} = I_{\text{original}} + \lambda \cdot I_{\text{laplacian}}$$

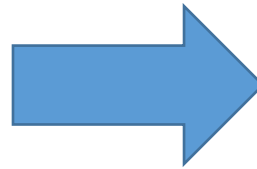
where λ is a scaling factor (typically 1).



Adding the image to the Laplacian restored the overall gray level variations in the image, with the Laplacian increasing the contrast at the locations of gray-level discontinuities.

The net result is an image in which small details were enhanced and the background tonality was perfectly preserved.

Original
Image



Enhanced
Image

7.7 Unsharp Masking and Highboost Filtering

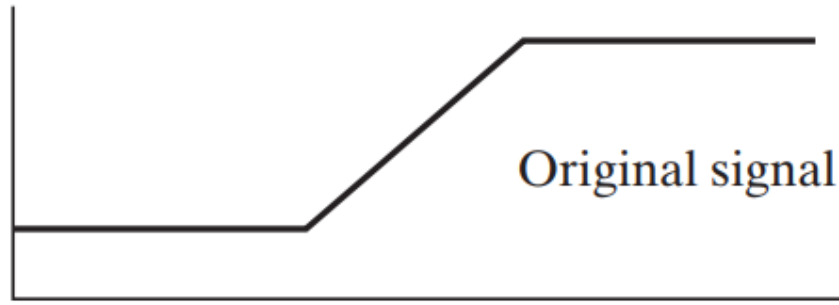
- ❑ Blur the original image.
- ❑ Subtract the blurred image from the original (the resulting difference is called the mask.)
- ❑ Add the mask to the original.

$$g_{\text{mask}}(x, y) = f(x, y) - \bar{f}(x, y)$$

$$g(x, y) = f(x, y) + k * g_{\text{mask}}(x, y)$$

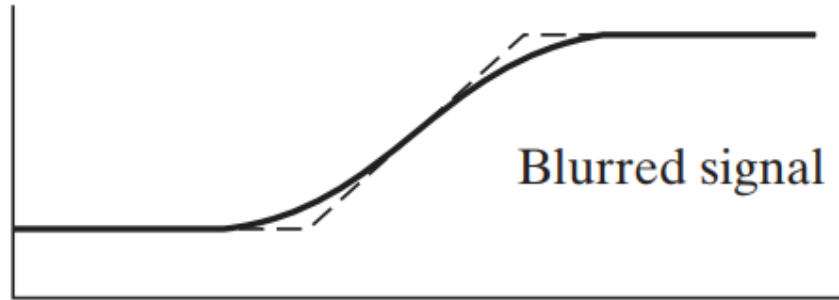
When $k=1$, we have unsharp masking, as defined above. When $k>1$, the process is referred to as highboost filtering. Choosing $k<1$ de-emphasizes the contribution of the unsharp mask.

(a)



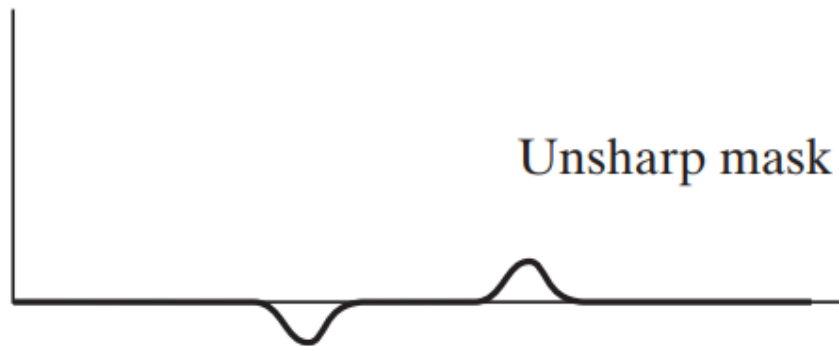
Original signal

(b)



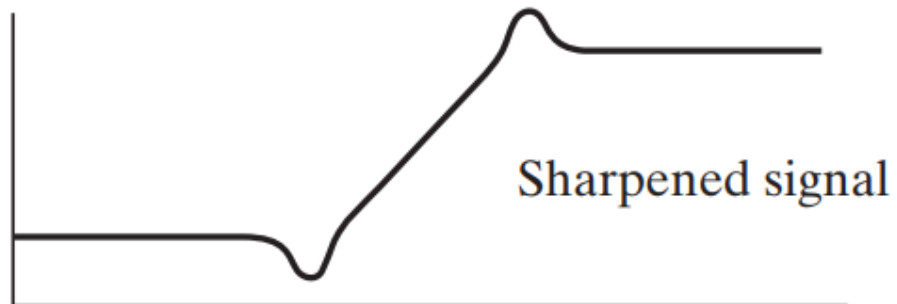
Blurred signal with original shown dashed for reference.

(c)



Unsharp mask

(d)



Sharpened signal, obtained by adding (c) to (a).



(a) Original image.



(b) Result of blurring with a Gaussian filter.



(c) Unsharp mask (Original - Blurred)



(d) Result of using unsharp masking ($k=1$)



(e) Result of using highboost filtering ($k=4.5$)

7.8 Using First-Order Derivatives for (Nonlinear) Image Sharpening—The Gradient

First derivatives in image processing are implemented using the magnitude of the gradient.

For a function f , the gradient of f at coordinates (x, y) is defined as the two-dimensional column vector

$$\nabla f \equiv \text{grad}(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

This vector has the important geometrical property that it points in the direction of the greatest rate of change of f at location (x, y) .

The magnitude (length) of vector ∇f denoted as $M(x, y)$, where

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

is the value at (x, y) of the rate of change in the direction of the gradient vector.

Note that $M(x, y)$ is an image of the same size as the original, created when x and y are allowed to vary over all pixel locations in f .

In image processing, a gradient represents the rate of intensity change (brightness variation) at each pixel.

Steps for Image Sharpening Using Gradients

Compute the Image Gradient

Apply first-order derivative filters to detect intensity changes. The gradient in both directions is computed as:

$$G_x = \frac{\partial I}{\partial x}, \quad G_y = \frac{\partial I}{\partial y}$$

Compute Gradient Magnitude

Combine the gradients to get the overall edge strength: $G = \sqrt{G_x^2 + G_y^2}$

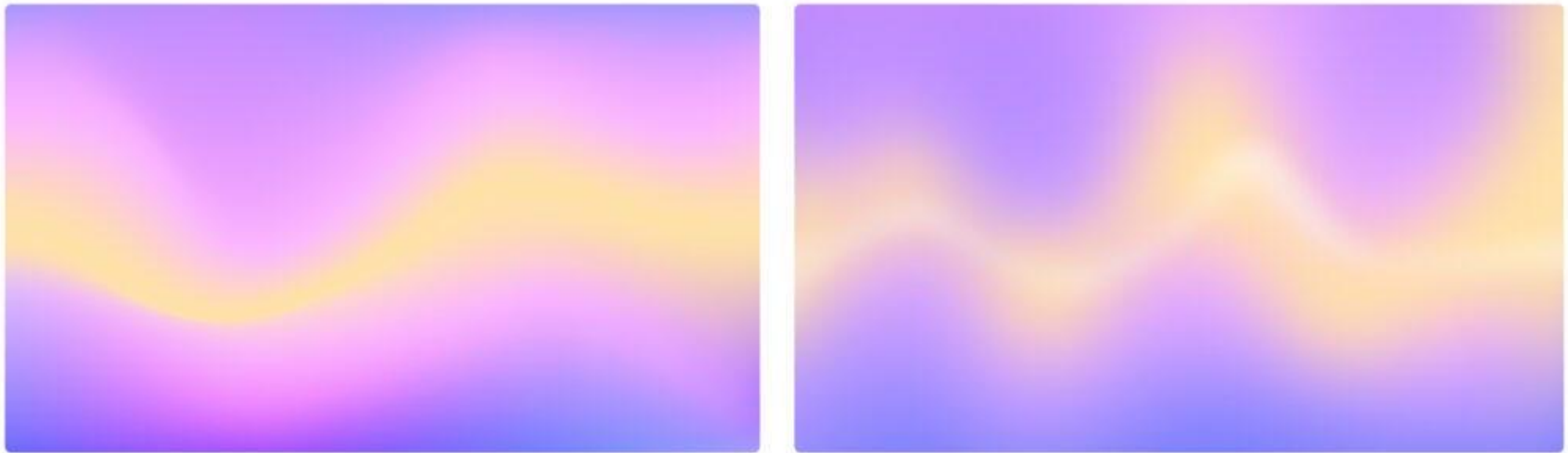
This highlights edges where intensity changes significantly.

Enhance the Image Using the Gradient

Add the gradient magnitude to the original image:

$$I_{\text{sharpened}} = I_{\text{original}} + \lambda G$$

where λ controls sharpening intensity.



Disadvantages of Gradients

- ❑ Because the components of the gradient vector are derivatives, they are linear operators.

$$\nabla f \equiv \text{grad}(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

- ❑ However, the magnitude of this vector is not because of the squaring and square root operations.

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

- ❑ On the other hand, the partial derivatives are not rotation invariant (isotropic), but the magnitude of the gradient vector is.

Solutions

- ✓ In some implementations, it is more suitable computationally to approximate the squares and square root operations by **absolute values**:

$$M(x, y) \approx |g_x| + |g_y|$$

Example: A region of an image (the z s are intensity values).

The center point z_5 , denotes $f(x, y)$ at an arbitrary location, (x, y) ;

z_1 denotes $f(x - 1, y - 1)$ and so on

The simplest approximations to a first-order derivative that satisfy the conditions stated in that section are

$$g_x = (z_8 - z_5)$$

$$g_y = (z_6 - z_5)$$

use cross differences

$$g_x = (z_9 - z_5) \quad \text{and} \quad g_y = (z_8 - z_6)$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

❑ If we use Eqs. $M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$

$$g_x = (z_9 - z_5) \quad \text{and} \quad g_y = (z_8 - z_6)$$

we compute the gradient image as

$$M(x, y) = \left[(z_9 - z_5)^2 + (z_8 - z_6)^2 \right]^{1/2}$$

✓ If we use Eqs. $M(x, y) \approx |g_x| + |g_y|$

$$g_x = (z_9 - z_5) \quad \text{and} \quad g_y = (z_8 - z_6)$$

we compute the gradient image as

$$M(x, y) \approx |z_9 - z_5| + |z_8 - z_6|$$

-1	0	0	-1
0	1	1	0

These masks are referred to as the Roberts cross-gradient operators

- ❑ Masks of even sizes are awkward to implement because they do not have a center of symmetry.

The smallest filter masks in which we are interested are of size 3 X 3

Approximations g_x and g_y using a 3×3 neighborhood centered on are as follows:

$$g_x = \frac{\partial f}{\partial x} = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$g_y = \frac{\partial f}{\partial y} = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

These equations can be implemented using the masks

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

After computing the partial derivatives with these masks, we obtain the magnitude of the gradient as before.

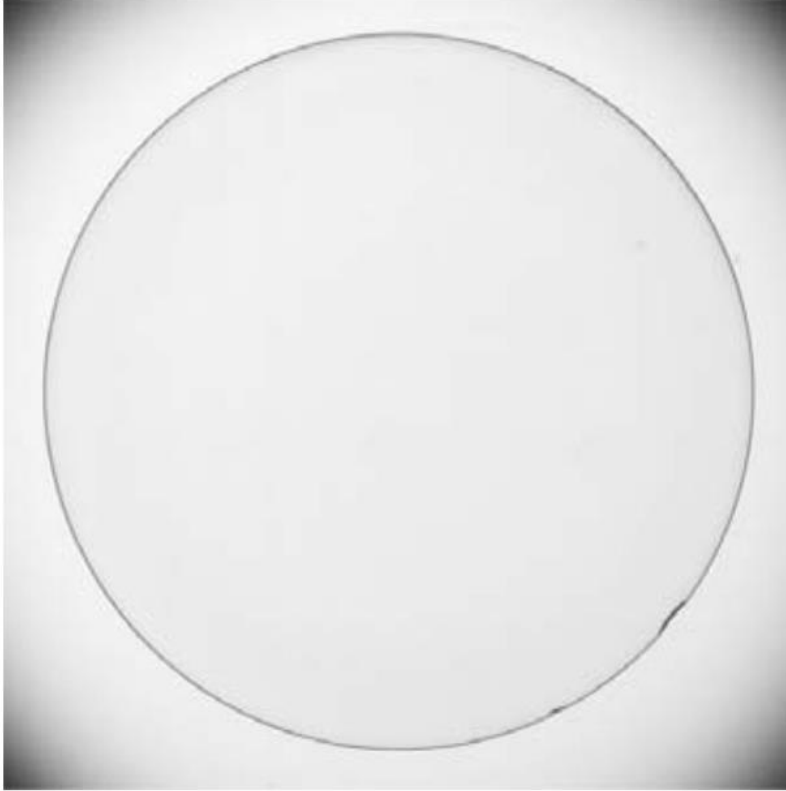
$$M(x, y) \approx |(z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)| \\ + |(z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)|$$

The masks in Figs. are called the **Sobel operators**.

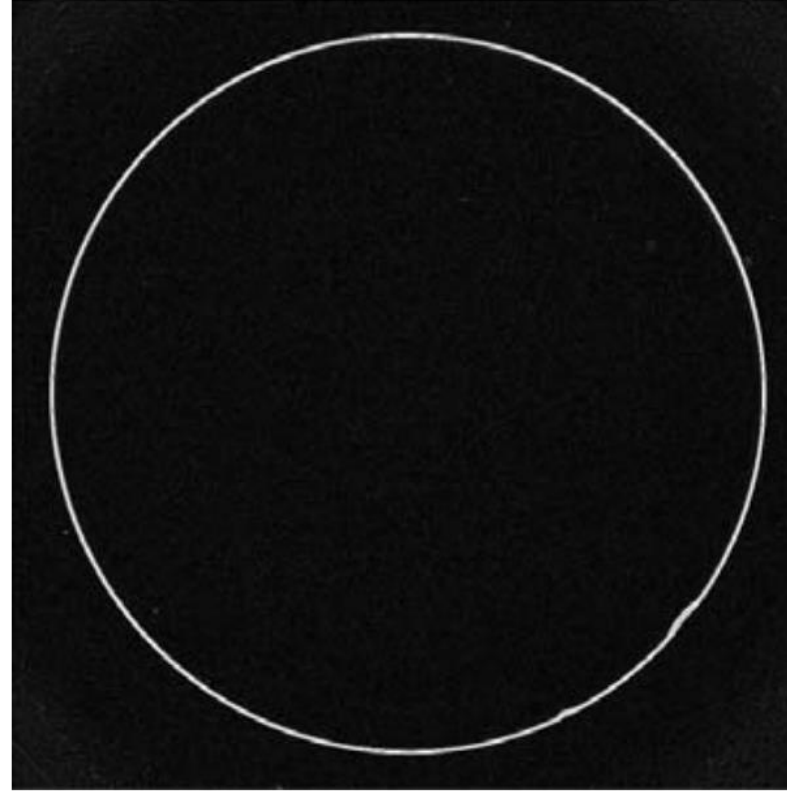
-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

The idea behind using a weight value of 2 in the center coefficient is to achieve some smoothing by giving more importance to the center point

Optical image
of contact lens



Sobel
gradient.



The edge defects also are quite visible in this image, but with the added advantage that constant or slowly varying shades of gray have been eliminated, thus simplifying considerably the computational task required for automated inspection.

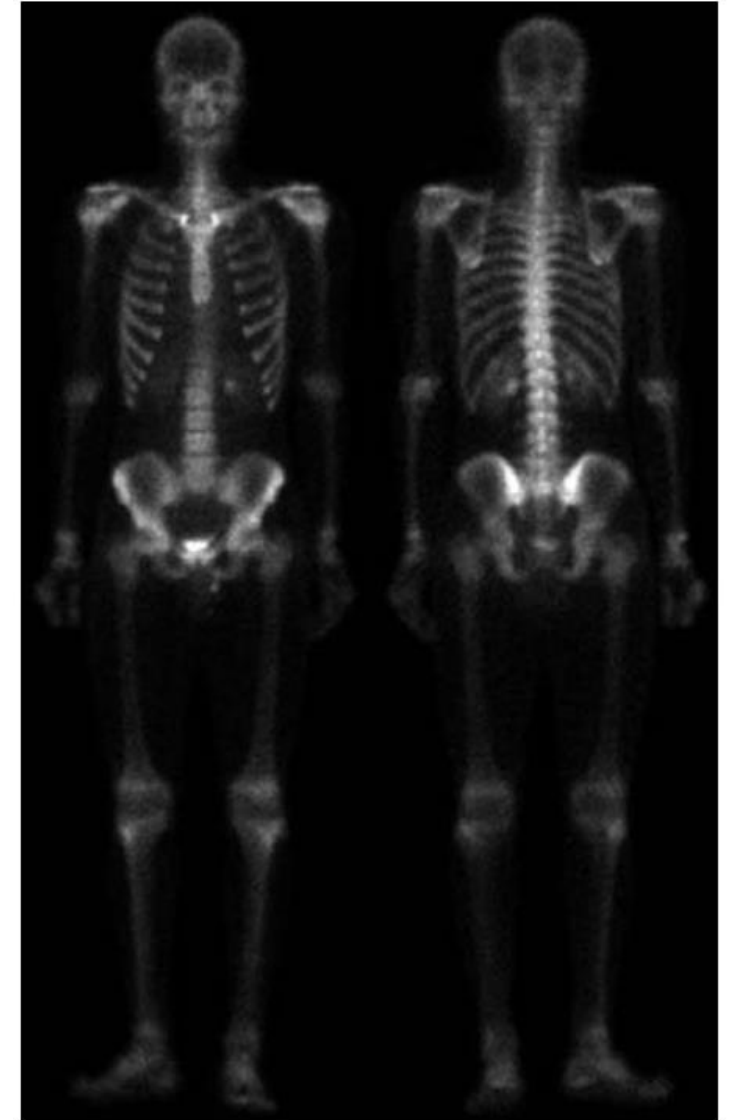
7.9 Combining Spatial Enhancement Methods

The image in Fig. is a nuclear whole body bone scan, used to detect diseases such as bone infection and tumors.

Goal: Our objective is to enhance this image by sharpening it and by bringing out more of the skeletal detail.

Challenges: The narrow dynamic range of the intensity levels and high noise content make this image difficult to enhance.

Way out: The strategy we will follow is to utilize the Laplacian to highlight fine detail, and the gradient to enhance prominent edges.

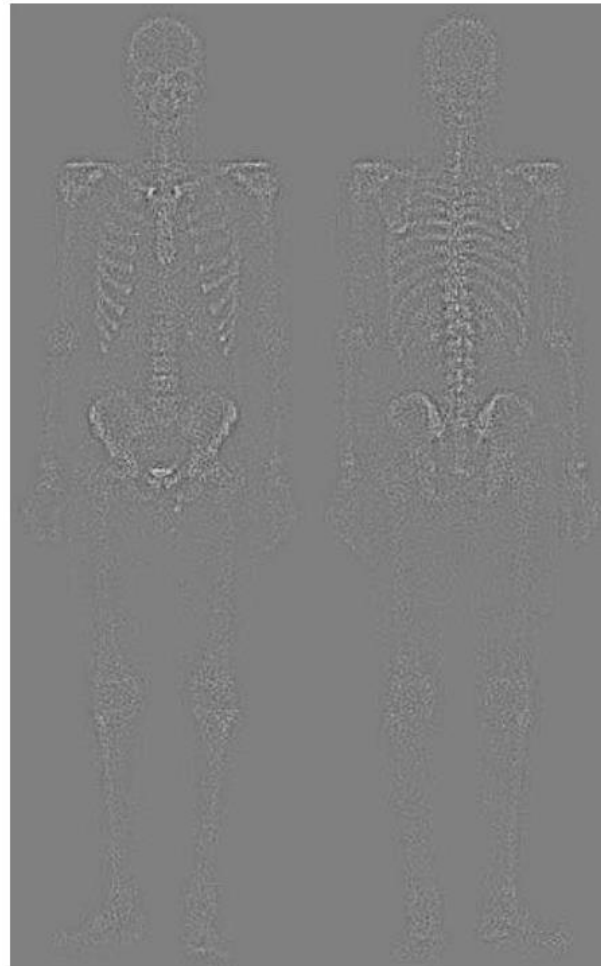


(a)

7.9.1 Utilize the Laplacian to highlight fine detail

Figure shows the Laplacian of the original image, obtained using the filter

Laplacian with
scaling

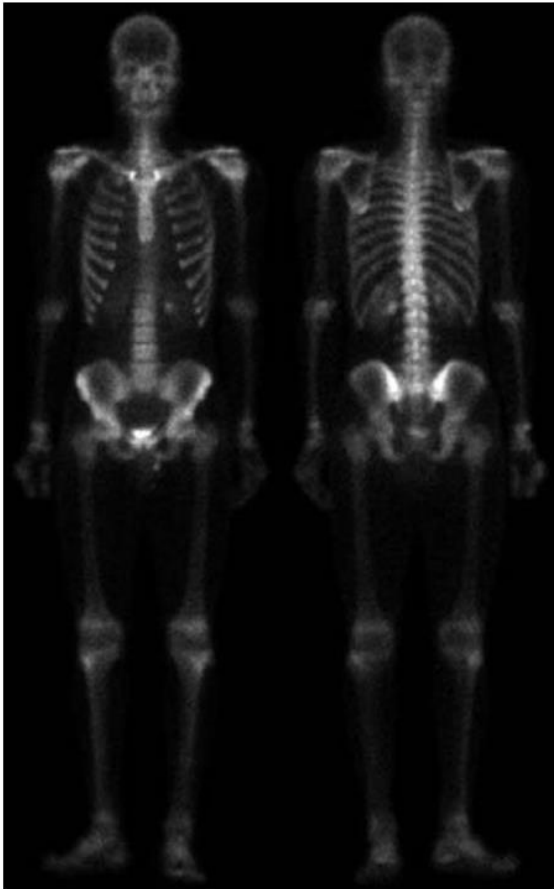


-1	-1	-1
-1	8	-1
-1	-1	-1

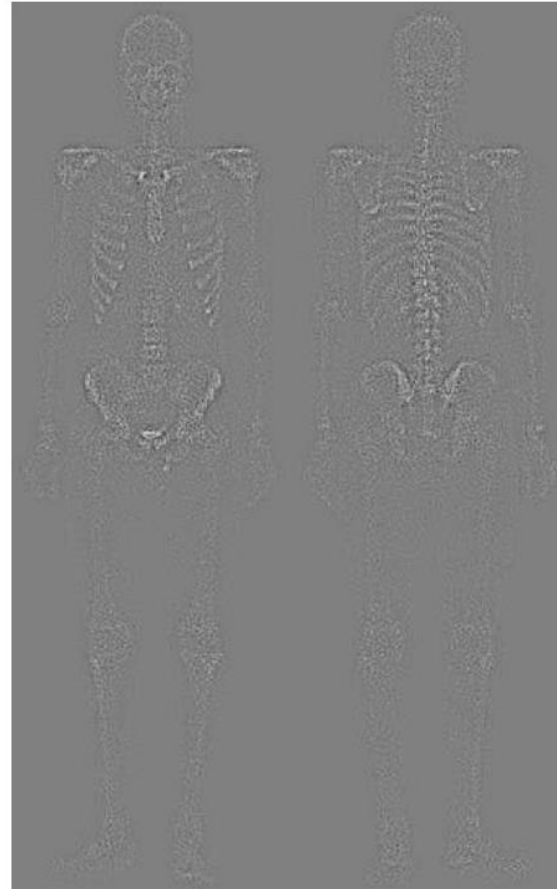
(b)

We can obtain a sharpened image at this point simply by adding (a) and (b)

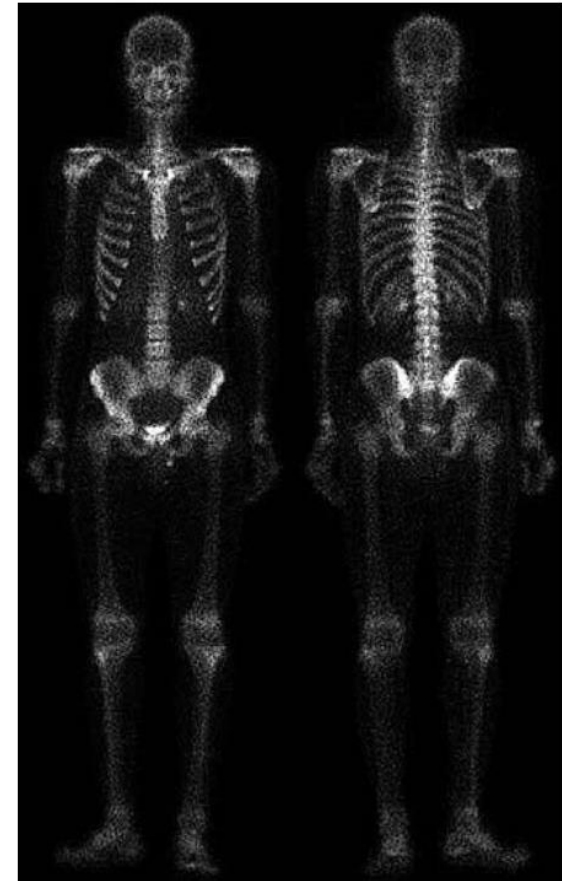
$$g(x, y) = \begin{cases} f(x, y) - \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is negative} \\ f(x, y) + \nabla^2 f(x, y) & \text{if the center coefficient of the} \\ & \text{Laplacian mask is positive.} \end{cases}$$



+



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Way to reduce the noise?

To reduce the noise is to use a median filter.

However, median filtering is a nonlinear process capable of removing image features.

This is unacceptable in medical image processing.

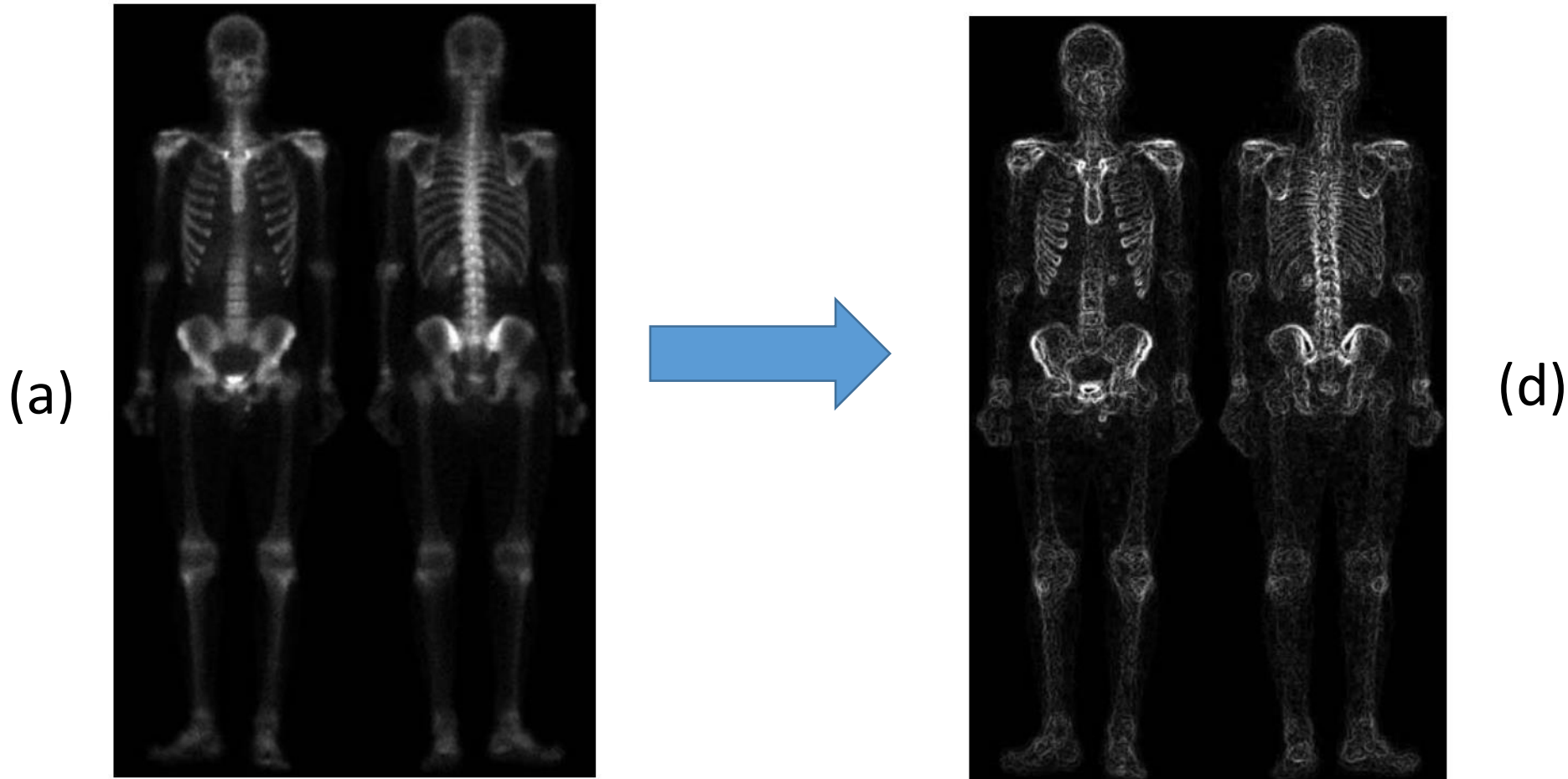
(c)



7.9.2 Gradient to enhance prominent edges

Figure shows the Sobel gradient of the original image, computed using Eq.

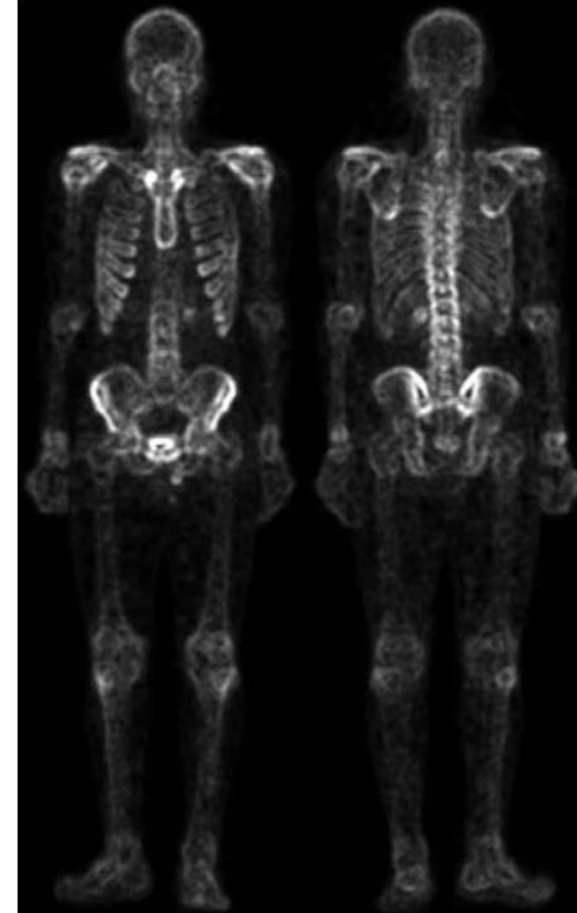
$$M(x, y) \approx |g_x| + |g_y|$$



Sobel image smoothed with a 5 X 5 averaging filter.



(d)



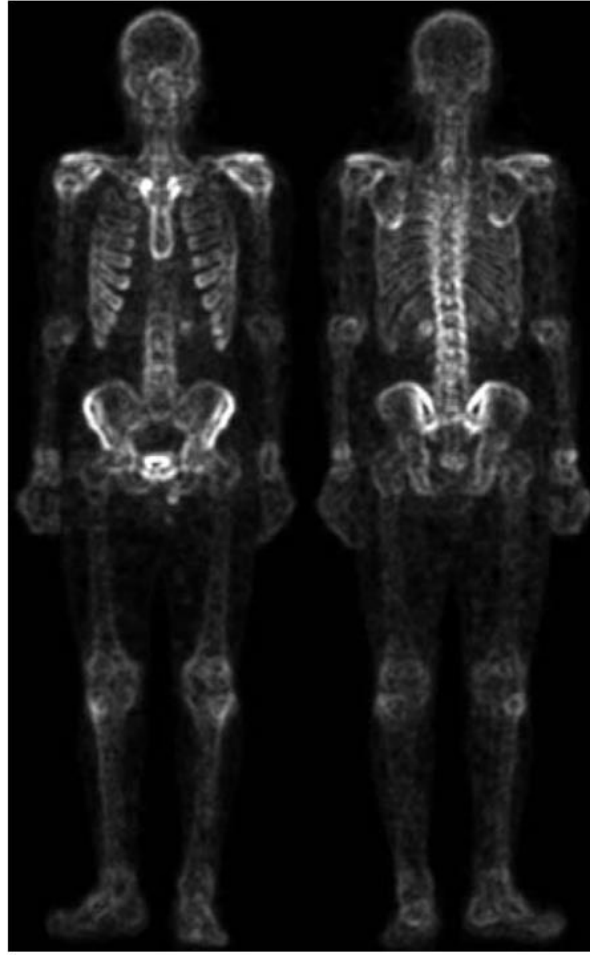
(e)

Mask image (f) formed by the product of (c) and (e).



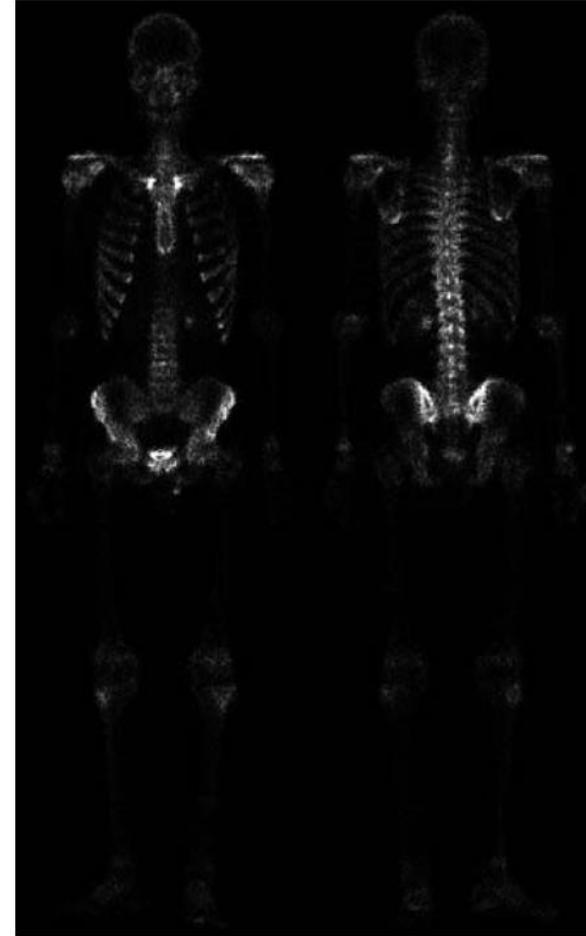
(c)

X



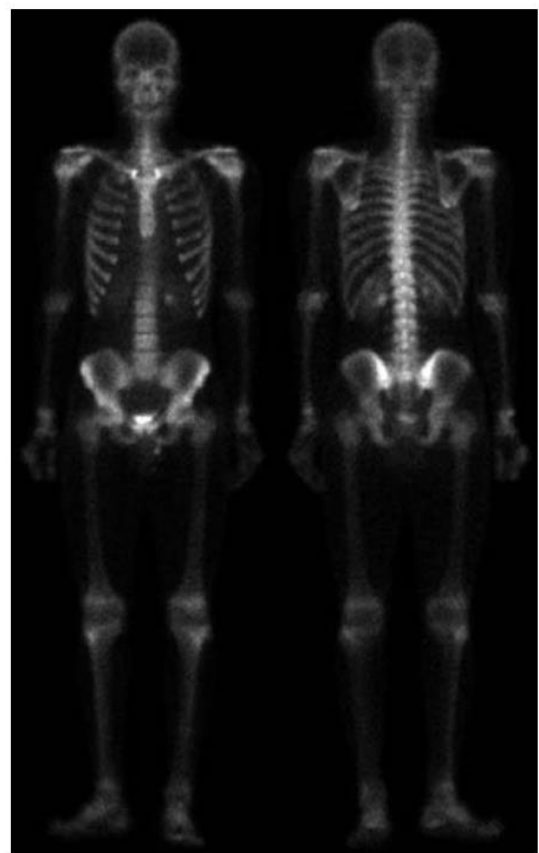
(e)

=



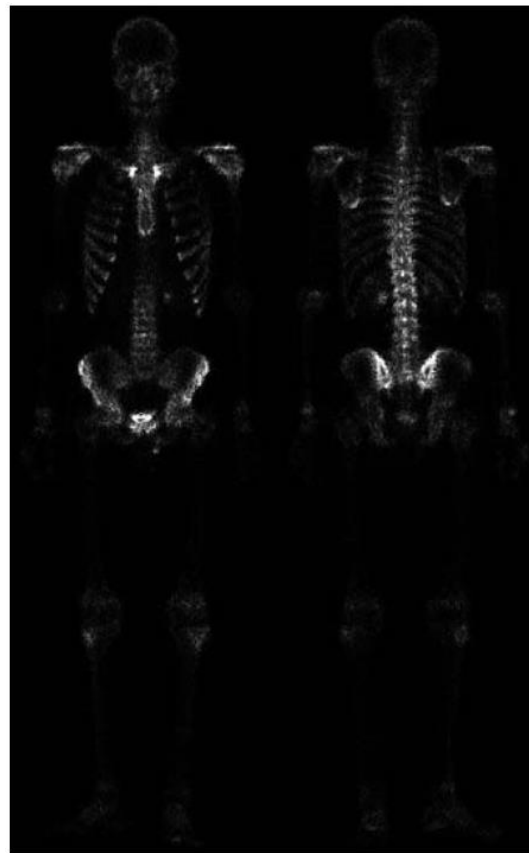
(f) Mask Image

Sharpened image (g) obtained by the sum of (a) and (f)



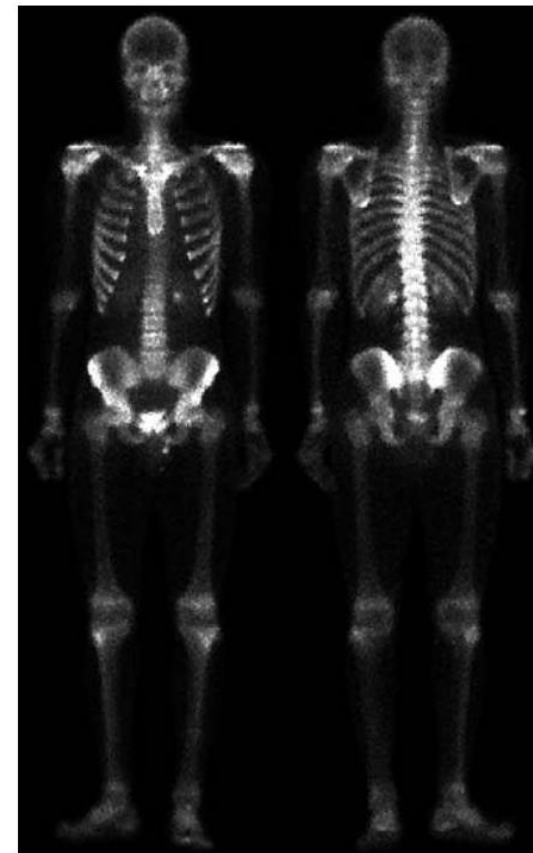
(a) Original image

+



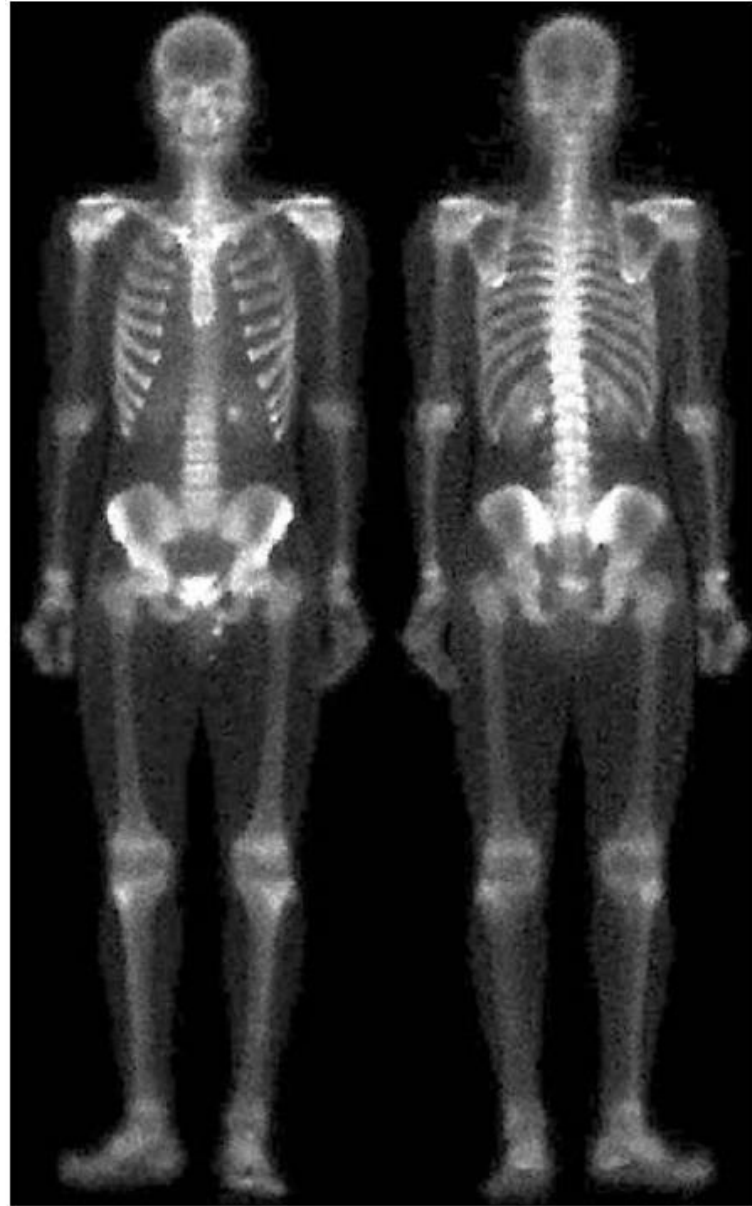
(f) Mask Image

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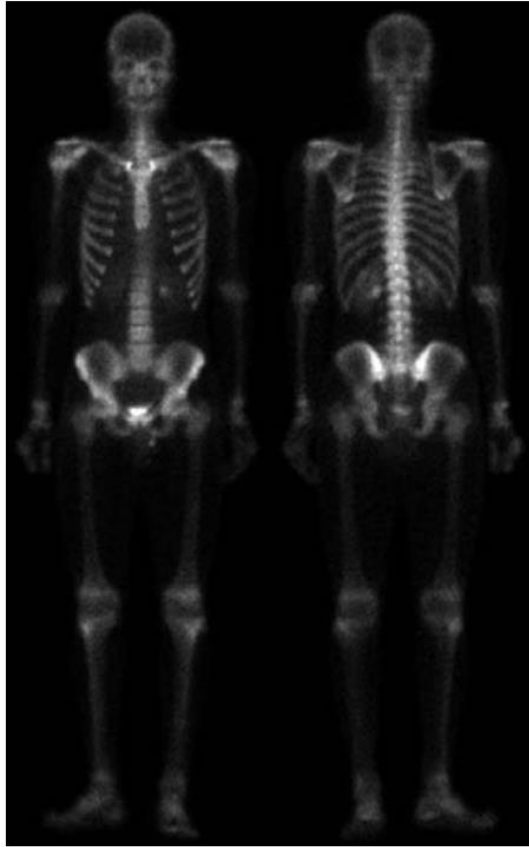
(g) Sharpened image

Final result (h) obtained by applying a power law transformation to (g).

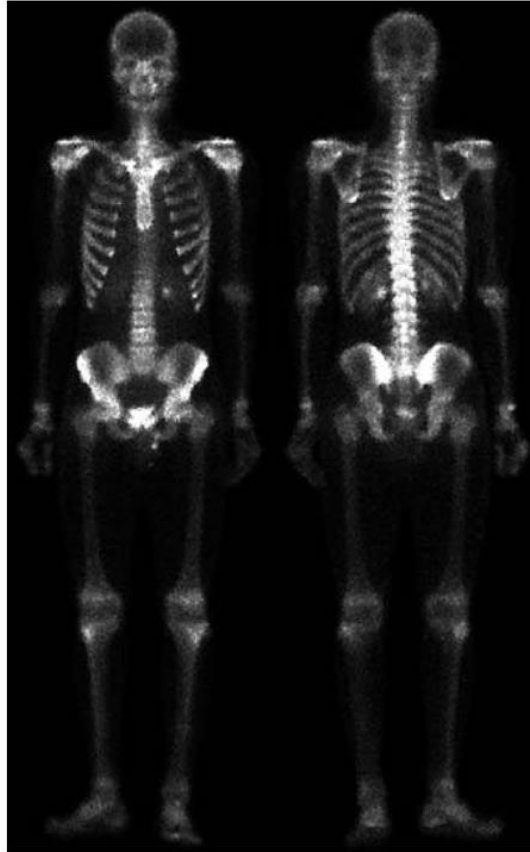


(h)

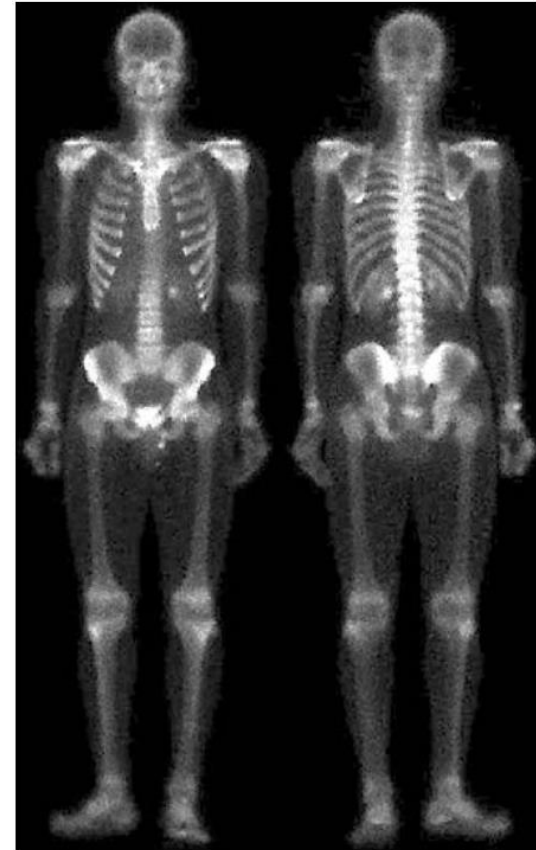
Compare (g) and (h) with (a)



(a) original



(g) Sharpened image



(h) Final enhanced Image